

Micro Energy Harvesting 2025

Exercise 3: Vibrational Energy Harvesting

Task 1: Quality Factor

The quality factor Q of the harmonic oscillator is also defined as the following ratio:

$$Q = 2\pi \frac{\text{energy stored}}{\text{av. energy lost per period}} = 2\pi \frac{W}{P \cdot T} = \frac{W}{P/\omega_0}$$

Derive the quality factor from the kinetic and potential energy:

$$W = W_{Kin} + W_{Pot}, \quad W_{Kin} = \frac{1}{2} m \dot{x}^2, \quad W_{Pot} = \frac{1}{2} k x^2$$

using the equation of motion for the damped harmonic oscillation:

$$x = x_0 \cdot \underbrace{e^{-\frac{t}{\tau}}}_{const.} \cdot \cos(\omega_0 \cdot t); \quad \tau = \frac{1}{\zeta \cdot \omega_0}; \quad \zeta \ll 1$$

Hint: Assume that the exponential term is virtually constant for the time span of one periode.

Solution:

First, we determine the velocity by building the temporal derivative of the displacement. Here we use the hint at treat the exponential term as constant. This keeps the solution much simpler.

$$\dot{x} = -\omega_0 \cdot x_0 \cdot \underbrace{e^{-\frac{t}{\tau}}}_{const.} \cdot \sin(\omega_0 \cdot t) \quad \text{eq. 1}$$

From that we can easily determine the kinetic and potential energy as well as their sum, the total Energy W .

$$\begin{aligned} W &= W_{Kin} + W_{Pot} = \\ &= \frac{1}{2} m \omega_0^2 x_0^2 \cdot e^{-\frac{2t}{\tau}} \cdot \sin^2(\omega_0 \cdot t) + \frac{1}{2} m k x_0^2 \cdot e^{-\frac{2t}{\tau}} \cdot \cos^2(\omega_0 \cdot t) = \\ &= \frac{1}{2} m \omega_0^2 x_0^2 \cdot e^{-\frac{2t}{\tau}} \end{aligned} \quad \text{eq. 2}$$

The power lost by the oscillator is the change in total energy per time

$$P = -\frac{dW}{dt} = \frac{2}{\tau} \frac{1}{2} m \omega_0^2 x_0^2 \cdot e^{-\frac{2t}{\tau}} \quad \text{eq. 3}$$

With eq. 2 and eq.3 we can simply determine the quality factor Q

$$Q = \frac{W}{P/\omega_0} = \frac{\frac{1}{2} m \omega_0^2 x_0^2 \cdot e^{-\frac{t}{2\tau}}}{\frac{2}{\tau} \frac{1}{2} m \omega_0^2 x_0^2 \cdot e^{-\frac{t}{2\tau}} \cdot \frac{1}{\omega_0}} = \frac{\tau \omega_0}{2} = \frac{1}{2\zeta} \quad \text{eq. 4}$$

Task 2: Radio protocol

Sending a radio protocol in a tunnel requires an assumed energy of $E_{\text{send}} = 200 \mu\text{J}$ (EnOcean STM300). Calculate the energy contained in the mass of $m = 1 \text{ g}$ with a resonance frequency of $f = 70 \text{ Hz}$ and an oscillation amplitude of $A_0 = 3 \text{ mm}$. How long would the energy last at a transmission frequency of $0,5 \text{ Hz}$? Assume that you can use 100% of the stored energy.

$$\omega_0 = 2\pi f = 439,823 \text{ Hz}$$

$$k = \omega_0^2 \cdot m = 193,4 \frac{\text{kg}}{\text{s}^2}$$

$$E = \frac{1}{2} k A_0^2 = 0,000873 \text{ J} = 873 \mu\text{J}$$

$$\text{Number of Radio protocols: } n = \frac{873 \mu\text{J}}{200 \mu\text{J}} = 4, \text{ Remainder } 73 \mu\text{J}$$

$$\text{Duration of service: } t = n/f = 4\text{s}/0,5 = \underline{\underline{8 \text{ s}}}$$

Task 3: Harvester Design

You want to design a harvester. First you measure the ambient vibration. You find the dominant vibration modes: 100 Hz with 50 μm vibration amplitude, 200 Hz with 20 μm vibration amplitude and 2.5 kHz with 5 μm vibration amplitude. You have the choice between two harvesters with the stiffnesses $K_1 = 1 \text{ mN}/\mu\text{m}$ and $K_2 = 4 \text{ mN}/\mu\text{m}$. The parasitic damping constants are $c_{\text{dis}1} = 10 \text{ g/s}$ and $c_{\text{dis}2} = 200 \text{ g/s}$. The generator damping of harvester 1 can be chosen arbitrarily, while $c_{\text{harv}2}$ can be chosen between $0.05 \text{ VAs}^2/\text{m}^2$ and $0.15 \text{ VAs}^2/\text{m}^2$.

- Choose the most suited set-up (harvester type 1 or 2, mass, c_{harv}). Justify your choice!
- What is the average power available for the application, if you assume a rectifier with 50% losses.
- Temperature changes lead to changes in material parameters and consequently resonance frequency. Which type of harvester is better in an environment with slightly varying temperature?

Solution**a) Best Set-up**

- Find the appropriate mass m_{ij} for each configuration (ij with i=1,2 referring to the two harvesters and j=1,2,3 referring to the three vibration modes) to achieve the

resonance frequency from $\omega_{0ij} = \sqrt{\frac{K_i}{m_{ij}}} = 2\pi \cdot f_j$.

j	1	2	3
f_j [Hz]	100	200	2500
x_j [m]	$50 \cdot 10^{-6}$	$20 \cdot 10^{-6}$	$5 \cdot 10^{-6}$
K_1 [N/m]	1000	1000	1000
K_2 [N/m]	4000	4000	4000
$m_{1,j}$ [kg]	0.0025	0.00063	$4.1 \cdot 10^{-6}$
$m_{2,j}$ [kg]	0.01	0.0025	$1.6 \cdot 10^{-5}$

- Calculate the maximum achievable power for each configuration from

$$P_{el,max} = \frac{1}{8} \cdot \frac{1}{c_{\text{dis}}} \cdot m^2 \cdot \omega_0^4 \cdot x_{\text{vib}}^2$$

$c_{\text{dis}1}$ [kg/s]	0,01	0,01	0,01
$c_{\text{dis}2}$ [kg/s]	0,2	0,2	0,2
$P_{\text{max}1}$ [W]	0,031	0,02	0,00031
$P_{\text{max}2}$ [W]	0,025	0,016	0,00025

- Check if you can match the load, if not recalculate the power from

$$P_{el}(c_{\text{harv}}) = \frac{1}{2} \cdot \frac{c_{\text{harv}}}{(c_{\text{harv}} + c_{\text{dis}})^2} \cdot m^2 \cdot \omega_0^4 \cdot x_{\text{vib}}^2$$

$P_{\text{max}1}$ [W]	0,031	0,02	0,00031
$P_{\text{max}2}$ [W]	0,0244	0,0157	0,000244

Harvester 1 optimized for the frequency of 100 Hz is the best solution, because it delivers the highest output power of 31 mW. This is surprising. The most energy is stored in the vibration at 2.5 kHz due to its high frequency. Unfortunately only very little mass can be added for both harvesters at these high frequencies.

The highest mass can be added to harvester 2 at 100 Hz, but it is unfortunately a highly dampened harvester. This is why harvester 1 at 100 Hz finally wins.

b) Available power?

Its just the output power calculated in a) times the rectifier efficiency of 50%, i.e. 15 mW

c) Best harvester for changing temperature?

Suddenly harvester 2 at 100 Hz wins. It delivers only 12.2 mW (50% of 24.4mW) compared to 15 mW from harvester 1, but its bandwidth of $\Delta\omega = \frac{\omega_0}{Q} = 5.5 \text{ Hz}$ is much higher than the 1.2 Hz of Harvester 1, i.e. it will harvest more energy compered to harvester 1 if the resonance frequency is off by about 1 Hz or more.

Task 4: Piezoelectric Harvester Design

You want to design a piezoelectric harvester for a resonance frequency of 100 Hz and a vibration amplitude of 10 μm . Its spring constant is $K = 1000 \text{ N/m}$, you can freely adapt its mass m . The damping constant is $c_{\text{mech}} = 0.02 \frac{\text{kg}}{\text{s}}$. Its electrical capacitance is $C = 100 \text{ nF}$ and its coupling coefficient is $\theta = 0.0008 \frac{\text{As}}{\text{m}}$. The coupling factor is $k = \sqrt{\frac{\theta^2}{C \cdot K}}$. The resonance frequency is slightly changed by the load resistance R according to

$$\omega_R \approx \omega_0 \sqrt{1 + \frac{k^2}{(1 - k^2)} \frac{1}{\left(\frac{RC\omega_0}{(1 - k^2)}\right)^2 + 1}}$$

Maximize the power output at resonance by adjusting m and R . What is resulting maximum output power and what is the power bandwidth?

Solution

- 1) The resonance frequency is near the natural frequency. Therefore, we adapt the mass by preliminary setting the natural frequency to 100 Hz

$$2 \cdot \pi \cdot f_0 = \sqrt{\frac{K}{m}} \Rightarrow m = \frac{K}{(2 \cdot \pi \cdot f_0)^2} = 0.00253 \text{ kg} = 0.56 \text{ mW}$$

- 2) Check for the type of optimization :

$$k = \sqrt{\frac{\theta^2}{C \cdot K}} = 0.08 \Rightarrow \frac{c_{\text{mech}}(1 - k^2)}{k^2 m \omega_0} = 1.96 > \frac{1}{2}$$

This means that you cannot achieve $c_{\text{mech}} = c_{\text{el}}$ and you have to maximize c_{el} .

- 3) Calculate optimum resistance

$$R_{\text{opt}} = \frac{1 - k^2}{C \omega_0} = 15.9 \text{ k}\Omega$$

- 4) Calculate resonance frequency

$$\omega_R \approx \omega_0 \sqrt{1 + \frac{k^2}{(1 - k^2)} \frac{1}{\left(\frac{RC\omega_0}{(1 - k^2)}\right)^2 + 1}} = \omega_0 \sqrt{1 + \frac{k^2}{(1 - k^2)} \frac{1}{2}} = 1.00002 \cdot \omega_0$$

This means that there is virtually no difference between resonance frequency and natural frequency . There is no need to change the mass determined earlier.

- 5) Calculate the quality factor

$$Q = \frac{m\omega_0}{c_{\text{dis}} + c_{\text{el}}};$$

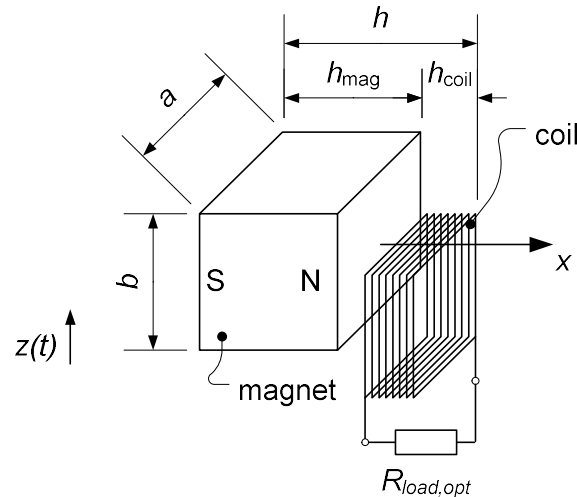
$$\begin{aligned} c_{\text{el}} &= \frac{mk^2 RC \omega_0^2}{(RC \omega_0)^2 + (1 - k^2)^2} = m\omega_0 \cdot \frac{k^2}{1 - k^2} \cdot \frac{\frac{RC \omega_0}{1 - k^2}}{\left(\frac{RC \omega_0}{1 - k^2}\right)^2 + 1} = m\omega_0 \cdot \frac{k^2}{1 - k^2} \cdot \frac{1}{2} \\ &= 0.01 \frac{\text{kg}}{\text{s}} \Rightarrow Q = \frac{m\omega_0}{c_{\text{dis}} + c_{\text{el}}} = 52.7 \end{aligned}$$

- 6) Calculate the maximum output power

$$P_{\text{el}}(c_{\text{harv}}) = \frac{1}{2} \cdot \frac{c_{\text{harv}}}{(c_{\text{harv}} + c_{\text{dis}})^2} \cdot m^2 \cdot \omega_0^4 \cdot x_{\text{vib}}^2 = \frac{1}{2} \cdot \frac{c_{\text{el}}}{c_{\text{mech}}^2} \cdot m^2 \cdot \omega_0^4 \cdot x_{\text{vib}}^2 = 557 \mu\text{W}$$

Task 5: Electrodynamic Harvester

For the following configuration of magnet (NdFeB, density: 7,819 g/cm³) and coil the output power for the optimal load resistance at resonance should be calculated. The overall size is fixed with $h = 15$ mm. The magnet should be cubic with an edge length of 10 mm.



Simplifying by assuming that no air gap between magnet and coil is required:

- How many turns N does the coil have with a wire diameter of $d_w = 40 \mu\text{m}$?
- The coil is rectangular as shown in the picture with edge length $l = a = 10$ mm. The wire has a resistance of 13.6 Ohm/m. What is the coil resistance?
- Calculate the optimum load resistance for a mechanical damping of $\zeta_m = 0,015$ and a resonant frequency of $f = 33$ Hz, coupling factor $k = 0,25$.
- Use the coupling factor shown below to calculate the output power of the system at resonance and $A_g = 1 \text{ m/s}^2$.

The coupling factor for this system is

$$k_t = N \cdot \frac{a}{h_{\text{coil}}} \int_0^{h_{\text{coil}}} B_x(x) dx$$

Determine with $\int_0^{h_{\text{coil}}} B_x(x) dx = 0.00132 \text{ T} \cdot \text{m}$

the real coupling factor k_t for this system with the remanence field of this magnet of $B_r = 1 \text{ T}$.

Solution

- 1) Coil optically as shown: turns on top of each other!

$$N = (h - h_{mag}) / d_D = 5000 \mu m / 40 \mu m = \underline{125}$$

- 2) Resistance from wire length

$$R_{Coil} = 4 \cdot l \cdot N \cdot 13,6 \Omega/m = 4 \cdot 10 \text{ mm} \cdot 125 \cdot 13,6 \Omega/m = 5000 \text{ mm} \cdot 13,6 \Omega/m = \underline{68 \Omega}$$

- 3)
- $m = 1 \text{ cm}^3 \cdot 7,819 \text{ g/cm}^3 = \underline{7,819 \text{ g}}$

$$\omega_{res} = f \cdot 2 \cdot \pi$$

$$k^2$$

$$R_{opt} = \frac{m \cdot \omega_{res} \cdot \zeta_m}{k^2} + R_0 = \underline{70,57 \Omega}$$

- 4)
- $c_{mech} = 2 \cdot m \cdot \omega_{res} \cdot \zeta_m$

$$k^2$$

$$c_{el} = \frac{R_0 + R_L}{k^2}$$

$$P_{max} = \frac{1}{2} \cdot \frac{c_{el}}{(c_{el} + c_{mec})^2} \cdot m^2 \cdot a_{vib}^2 \cdot \frac{R_L}{R_0 + R_L} = 13,35 \text{ mW}$$

- 5)
- $k_t = N \cdot \frac{a}{h_{coil}} \int_0^{h_{coil}} B_x(x) dx = 125 \cdot \frac{10 \text{ mm}}{5 \text{ mm}} \cdot 0,00132 \text{ T} \cdot m = \underline{0,33}$